

# BIOENERGETICS – PHOTOSYNTHESIS

KEY VOCABULARY

**Photosynthesis:** plants make glucose using light energy.

**Chlorophyll:** green pigment that absorbs light energy.

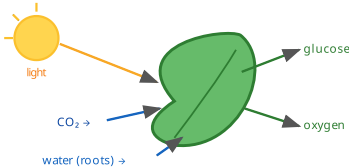
**Endothermic:** a reaction that takes in energy.

**Limiting factor:** the factor that stops the rate increasing further.

COMMON MISCONCEPTIONS

- Plants only respire at night.
- ✓ Plants respire all the time, day and night.
- Plants get food from the soil.
- ✓ Plants make glucose in their leaves.
- Photosynthesis releases energy.
- ✓ It is endothermic — it absorbs light energy.

INPUTS & OUTPUTS



carbon dioxide + water → **glucose** + oxygen (light/chlorophyll).

1 WORD EQUATION

Complete the word equation for photosynthesis. [2 marks]

\_\_\_ + water → \_\_\_ + oxygen

carbon dioxide

oxygen

glucose

water

2 ENDOTHERMIC

Explain why photosynthesis is described as an endothermic reaction. [2 marks]

3 LIMITING FACTORS

State **three** factors that can limit the rate of photosynthesis. [3 marks]

4 INTERPRET A GRAPH

A graph of rate against light intensity levels off. Explain why. [2 marks]

5 USES OF GLUCOSE

Give **two** ways a plant uses the glucose it makes. [2 marks]

6 INVERSE SQUARE LAW

A lamp is moved from 10 cm to 20 cm from a plant. State what happens to the light intensity. [1 mark]